

Universidade Católica Portuguesa

Escola Superior de Biotecnologia

MSc Biomedical Engineering

Your Thesis Title Goes Here

Master's Thesis

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Porto, Portugal

Month 2027

Abstract

[Replace this placeholder: Write a concise abstract here. A strong scientific abstract usually contains the background/problem, objective, methods, key results, and conclusion.]

Keywords: *[Replace this placeholder: keyword 1; keyword 2; keyword 3; keyword 4; keyword 5]*

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List of Abbreviations

MRI Magnetic Resonance Imaging.

1. Introduction

1.1. Background

[Replace this placeholder: Introduce the scientific and clinical background. Define the problem and explain why it matters.]

A first citation can be written as follows [1]. The abbreviation Magnetic Resonance Imaging (MRI) is expanded automatically on first use.

1.2. Research Gap

[Replace this placeholder: Describe what is not yet sufficiently understood or solved.]

1.3. Aim and Research Questions

[Replace this placeholder: State the main aim and, where appropriate, explicit research questions or hypotheses.]

1.4. Structure of the Thesis

[Replace this placeholder: Briefly explain what each chapter contains.]

2. Materials and Methods

2.1. Dataset

[Replace this placeholder: Describe the dataset, inclusion/exclusion criteria, sample size, acquisition protocol, preprocessing, and relevant ethics/data-governance aspects.]

2.2. Image Preprocessing

[Replace this placeholder: Describe registration, skull stripping, normalization, segmentation, resampling, denoising, bias-field correction, or other preprocessing steps as applicable.]

2.3. Feature Extraction

[Replace this placeholder: Describe the features, extraction method, software, and parameters.]

2.4. Machine Learning / Deep Learning

[Replace this placeholder: Describe model architecture, train/validation/test split, preprocessing, class balancing, hyperparameter selection, training procedure, and reproducibility.]

2.5. Statistical Analysis

[Replace this placeholder: Describe statistical tests, assumptions, confidence intervals, effect sizes, and significance thresholds.]

2.6. Software and Reproducibility

The analysis was implemented in Python. Record the relevant package versions and the operating environment here.

3. Results

3.1. Cohort / Dataset Characteristics

[Replace this placeholder: Present descriptive statistics and dataset characteristics.]

3.2. Feature Results

[Replace this placeholder: Present extracted features and important observations.]

3.3. Model Performance

[Replace this placeholder: Report performance metrics with uncertainty and validation details.]

Table 3.1.: Example model-performance table. Replace all values before submission.

Model	Accuracy	Precision	Recall
Baseline	0.00	0.00	0.00
Model A	0.00	0.00	0.00
Model B	0.00	0.00	0.00

As shown in table 3.1, this is only a formatting example.

4. Discussion

4.1. Main Findings

[Replace this placeholder: Interpret the principal findings without repeating the Results chapter.]

4.2. Comparison with Previous Literature

[Replace this placeholder: Compare your findings with the relevant literature and explain similarities, differences, and plausible methodological reasons.]

4.3. Limitations

[Replace this placeholder: Discuss limitations in data, methodology, generalisability, statistical power, and potential bias.]

4.4. Implications and Future Work

[Replace this placeholder: Describe what the findings imply and what should be investigated next.]

5. Conclusion

[Replace this placeholder: State the main conclusion directly and concisely. Answer the research question without introducing new results or references.]

References

- [1] C. R. Jack, D. A. Bennett, K. Blennow, M. C. Carrillo, B. Dunn, S. B. Haeberlein, D. M. Holtzman, W. Jagust, F. Jessen, J. Karlawish, E. Liu, J. L. Molinuevo, T. Montine, C. Phelps, K. P. Rankin, C. C. Rowe, P. Scheltens, E. Siemers, H. M. Snyder, R. Sperling, and K. Yaffe. “NIA-AA Research Framework: Toward a biological definition of Alzheimer’s disease”. In: *Alzheimer’s & Dementia* 14.4 (2018), pp. 535–562. doi: 10.1016/j.jalz.2018.02.018.

A. Additional Material

[Replace this placeholder: Use appendices for detailed material that is useful for transparency but would interrupt the main narrative, such as extra tables, questionnaires, parameter lists, or code.]